

Semantic Theory 2026: Exercise 6

Due by: Wednesday, June 10 at 10:00 am (before class)

Question 1

A model structure for plural terms is a tuple $M = ((U_M, \leq), V_M)$, where $(U_M, \leq)$ is an atomic join semi-lattice with universe U_M and individual part relation $\leq$, and V_M is an interpretation function mapping elements of the logical language to elements of the universe.

Consider the model M_1 , where the universe U_{M_1} is generated by the following set of atoms: $\{a, b, j, m, s\}$.

- a. Assume that $\llbracket \text{John, Mary, and Bill sing} \rrbracket^{M_1} = 1$, $\llbracket \text{Albert sings} \rrbracket^{M_1} = 1$, and $\llbracket X \text{ sing}(s) \rrbracket^{M_1} = 0$ for all other individuals (and proper sums) X for the predicate $\text{sing} \in P_d$. Then:

$$V_{M_1}(\text{sing}) = ?$$

- b. Assume that $\llbracket \text{John and Mary meet} \rrbracket^{M_1} = 1$, $\llbracket \text{Albert and Sally meet} \rrbracket^{M_1} = 1$, $\llbracket \text{Bill and Mary meet} \rrbracket^{M_1} = 1$, and $\llbracket X \text{ meet} \rrbracket^{M_1} = 0$ for all other individuals (and proper sums) X for the predicate $\text{meet} \in P_c$. Then:

$$V_{M_1}(\text{meet}) = ?$$

Question 2

Consider the following sentence: $S = \text{“two students summarized three papers”}$

- a. How many readings does the sentence S have? List all possible readings in natural language.
- b. Translate each reading of S to the extended first-order logic for plural terms introduced in the lecture, which extends first-order logic with a $\oplus$ operator, a $\triangleleft$ operator, and variables $X, Y, Z, \dots$ ranging over proper sums: $X \oplus Y$ denotes the group consisting of X and Y , $\triangleleft$ denotes the part-of-relation.

You may (and should) also incorporate the function $N(X) = |\{y \mid \text{At}(y) \wedge y \triangleleft X\}|$ that takes a proper sum X and returns the number of atoms in X .